INCH-POUND

MIL-DTL-23020C
17 November 2005
SUPERSEDING
MIL-C-23020B (SHIPS)
12 April 1965

DETAIL SPECIFICATION

CABLE, COAXIAL (FOR SUBMARINE USE)

Inactive for new design after 16 June 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the specific requirements for coaxial cables intended for submarine applications.

1.2 Part or Identifying Number (PIN). The PIN consists of the "RG" number as specified on the associated specification sheet:

Example: RG-264C/U

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center Columbus, ATTN: VAI, P.O. Box 3990, Columbus OH 43218-3990 or by email to RFConnectors@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

ASMC N/A FSC 6145

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-17 - Cables, Radio Frequency, Flexible and Semirigid, General Specification, For.
MIL-DTL-915 - Cable, Cord and Wire, Electrical (Shipboard Use).

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the requirements cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article sample. When specified (see 6.3), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 General requirements. The requirements of MIL-DTL-17 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, the requirements of this specification shall govern.

3.4 Design and construction. The design, construction, and physical dimensions shall be as specified in the specification sheets.

3.5 Materials. The cable shall be constructed of material as specified in the specification sheet. When a specific material is not specified, it shall be of a type and quality to assure physical and electrical compatibility with the requirements of this specification. Approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.5.1 Waterproofing compounds. The cables procured in accordance with this specification shall be watertight along the axis of the cable to the extent specified. In order to meet this requirement, a water sealant, compound, or paste may be used as a filler, provided the cables so constructed meet all other requirements. The waterproofing procedure shall be such that the individual wires of the outer conductor (braid) are readily separable from the jacket, filler, and core, and the construction shall be such as to maintain ease of attachment to standard connectors without significant degradation of electrical characteristics as compared with similar assemblies using standard cables.

3.6 Dielectric. The dielectric shall make mechanically intimate contact with the inner conductor (or conductors) so that no air film, air bubbles, or air voids are present. There shall be no marks or imperfections on the core which would prevent proper sealing with an O-ring. The inner conductor shall not be off-center more than 10 percent of the core radius.

3.7 Jacket thickness. The minimum jacket thickness shall be as specified in the specification sheet.

3.8 Abrasion. When tested as specified in 4.4.3, the cable shall withstand the number of revolutions specified in the specification sheet prior to cut-through.

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3.9 Watertightness. When tested as specified in 4.4.2, the leakage in a 2-hour period shall not exceed the value specified in the specification sheet.

3.10 Dimensions. The manufacturer of cables to this specification shall provide the necessary equipment to monitor continually the physical dimensions of the cable core and overall diameter.

3.11 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.12 Marking. Shipment marking information shall be provided on both flanges of each reel in accordance with the contractor's commercial practice. The information shall include the following:

a. Cable description.
b. Federal stock number.
c. Number and size of conductors.
d. Quantity (footage of all lengths).
e. Contract or order number.
f. Contractor's name or manufacturer's Cage code.
g. Year of manufacture.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall consist of the tests of MIL-DTL-17 with the exceptions and additions specified in 4.4. Prior to beginning production, a sample shall be tested as specified and the sample shall be from production tools, and shall be a minimum of 250 feet (76.2 m) long.

4.3 Conformance inspection. Conformance inspection shall be in accordance with acceptance inspection of MIL-DTL-17.

4.3.1 Unit of product. For purposes of this specification, a unit of product shall be defined as 1000 feet (304.8 m) cable having the sample type designation.

4.4 Test procedures.

4.4.1 Cold working. A 20-foot (6.1 m) sample from each reel shall be conditioned for not less than 20 hours at -54° + 0-5°C. Immediately thereafter but while the specimen is still in the cold chamber at the conditioning temperature, the specimen shall be drawn at a rate of not less than 100 feet (30.5 m) per minute over a roller having a diameter of 12 inches (0.3 m), in such a manner that a "U" bend of 180 degrees is imposed progressively on the entire length. The specimen shall then be drawn in the reverse direction at the same rate. It shall be drawn alternately forward and backward, for a total of four times in each direction. The specimen shall then be removed from the cold chamber and visually examined for evidence of cracks or flaws in the dielectric material or jacket. A length of the cable employed in the cold working test shall then meet the requirements of 3.8.

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4.4.2 Hydrostatic test (see 3.9). The hydrostatic test shall be conducted in accordance with MIL-DTL-915, except that under the water-tightness test procedure specified in MIL-DTL-915, the methods of sealing and clamping the cable jacket shall be in accordance with the methods employed in the pressure proof fittings designated herein for the individual types. When the applicable fitting employs an "O" ring to seal around the cable core, the "O" ring shall not be installed during the hydrostatic test. The test shall be made at the pressure and rate of leakage specified on the specification sheet for a period of 2 hours. If the cable fails the hydrostatic test it shall be rejected. The sampling for this test shall be a 5-foot (1.5 m) piece of each manufacturer's length. Retesting will be permitted only when, in the judgment of the Government representative, the previous test was improperly conducted.

4.4.2.1 Fittings. Fittings for the hydrostatic test shall be in accordance with the drawings listed in section 2 and as specified on the specification sheets.

4.4.3 Abrasion (see 3.8). The abrasion test shall be in accordance with MIL-DTL-915.

4.5 Visual and mechanical examination. The diameters and out-of-roundness of the core and jacket shall be monitored on a continuous production basis by a suitable technique which will provide the diameter in two mutually perpendicular planes at any cross section. The dimensions shall be as specified and out-of-roundness (the difference in diameter dimension is mutually perpendicular planes at any cross section) shall not exceed 50 percent of the difference between the maximum and minimum diameters specified. The results shall be permanently recorded by a two-pen strip chart recorder or by another electronic or mechanical means to record the diameter variances as long as the contracting officer approves of the method and the method sensitivity is equal to or better than the two-pen strip chart method. The test data shall be furnished to the procuring activity under separate cover at the time of cable shipment. Each recording shall be identified as to the reel of cable to which it applies, and each recording shall be identified to within ± 2 feet (0.6 m) of the corresponding point on the cable being measured. The dimensional recordings shall be individually identified as to:

a. Contract.
b. Cable type.
c. Manufacturer.
d. Footage of corresponding cable.
e. Whether applicable to dielectric or jacket measurements.
f. Calibration.

4.5.1 The technique used (including detector, recorder, and associated units) shall meet the following:

a. Have a sensitivity to record accurately the cable diameter to the third decimal place 0.001 inch (0.02 mm) along the length of cable.
b. Have a response capable of accurately recording changes in diameter as specified in (a) at the speed (feet/minute) of cable production. Deviation occurring on a length 0.060 inch (1.5 mm) or greater shall be discernible.
c. The strip chart recorder pen traverse shall be large enough to distinguish easily changes in diameter of 0.001 inch (0.02 mm).

The ratio of cable production in feet/minute to chart travel in inches/minute shall be small enough to permit ease in comparing the individual traces from the two pens. A 20:1 ratio shall be considered maximum acceptable ratio.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military service or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 These cables are intended for use in submarine radio frequency applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Cable type required (see specification sheets).
c. Packaging as specified in 5.1.

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has previously been procured or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid prior Government approval is presently appropriate for the pending procurement.

6.4 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. Table I lists the Environmental Protection Agency (EPA) top seventeen hazardous materials targeted for major usage reduction. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (see 3.5).

TABLE I. EPA top seventeen hazardous materials.

Benzene	Dichloromethane	Tetrachloroethylene
Cadmium and Compounds	Lead and Compounds	Toluene
Carbon Tetrachloride	Mercury and Compounds	1,1,1 - Trichoroethane
Chloroform	Methyl Ethyl Ketone	Trichloroethylene
Chromium and Compounds	Methyl Isobutyl Ketone	Xylenes
Cyanide and Compounds	Nickel and Compounds	

6.5 Reference information. The following documents may be used with the cables associated with this specification:

REC49366 - MX-2327/U – Stuffing Tube for Use with MIL-C-17/215 Cable (RG-14A/U).
REB49336 - MX-2326/U – Stuffing Tube for Use with MIL-C-17/216 Cable (RG-17/U).
REB49397 - MX-2646/U – Stuffing Tube (RG-57/U).
44500-8-69 - Stuffing Tube.

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6.6 Subject term (key word) listing

Dielectric

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Navy – EC
DLA - CC

Preparing activity:
DLA – CC

(Project 6145-2390-000)

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